

Topic

- Turbulence propagation along field lines in stellarators (HSX)
 - Specifically, we will compare how turbulence events propagate along field lines in the quasi-helical configuration of HSX versus how they propagate along field lines in the mirror geometry configuration of HSX and we will do this comparison for multiple values of the parallel connection length

Reasons for doing it

- Turbulence limits plasma confinement time as a result of chaotic transport carrying heat and particles across field lines and mixing hotter core plasma with cooler edge plasma
- Additionally, the turbulent plasma creates fluctuating electric fields which cause $E \times B$ guiding center drift
- And, stellarators with their complicated geometries make it hard to model turbulence since turbulence in different regions of plasma can behave very differently
- Additionally, the fact that turbulence can spread along field lines can mean that stable areas of plasma can become unstable and turbulence at a single source can affect the entire plasma
- The HSX stellarator specifically has a different geometry from many other stellarators in that it is quasi-helically symmetric but also has magnets installed that when turned on can break the quasi-helical symmetry and turn it into a magnetic mirror configuration
- This allows us to do a direct comparison of the way turbulence propagation changes as a result of the mirror configuration since it is the same device with the exact same plasma but all that is changing is whether or not the mirror magnets are turned on (in other words, studying HSX allows for a cleaner comparison without having to worry about other effects caused by differences in device design if two different devices were used to do this comparison)

Description of Project

- We will first use SIMSOPT to trace the field lines of the HSX stellarator in both its quasi-helical symmetry configuration and its mirror configuration and extract how the magnetic field strength varies along each field line in both configurations in a format that we can use as an input for the rest of our code
- We will then convert all of our 3D field line data into 1D field strength data as a function of distance along a particular field line (2D slab model)
- We will then build a numerical solver in Python for the 1D parallel transport equation (diffusion equation) for each configuration so that we can see how the plasma evolves over time along the now 1D field lines in our 2D slab model
- We will then run both of these solvers for many different parallel connection lengths and record how quickly the turbulence fluctuations evolve and decay as well as how far they propagate before they decay both among different parallel connection lengths for the same configuration and among the same parallel connection lengths for the different configurations

Tools we'll use

- SIMSOPT
- VMEC for HSX (if we can get it)
- Our own solver for a 1D diffusion equation in Python

Faculty Advisor

- We are currently waiting to hear back from Dr. Heinke Frerichs

Referenced Literature

- None yet although we hope to get some advice as to what we need to read up more on from Dr. Frerichs when we meet him

AI Disclosure

- Because we have so little knowledge of what is going on in plasma physics, we used AI to help us narrow down some ideas for potential topics into a single topic and a more specific plan as to what would be involved in doing a project like this so that we could assess whether or not it seemed possible in the time we have.
- Everything we have written in this document is our own work, however we had to learn a lot in order to put this together so we used AI for a lot of our learning (defining terms, learning what research exists, what different plasma tools do, etc.)